Anonymous Credential in Sphinx Packets

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August 6, 2026

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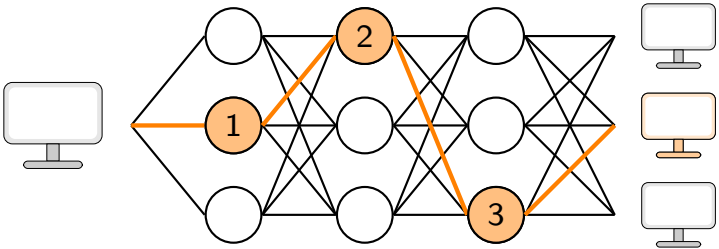
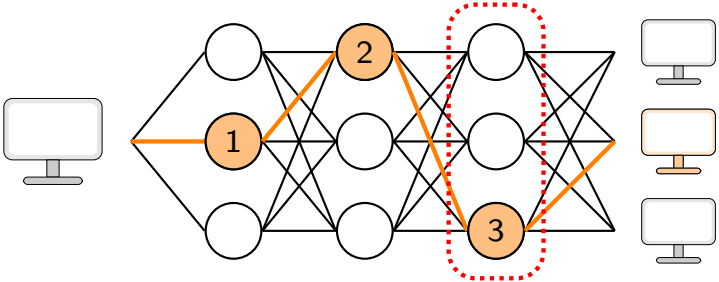
My Research - Credentials

Motivation

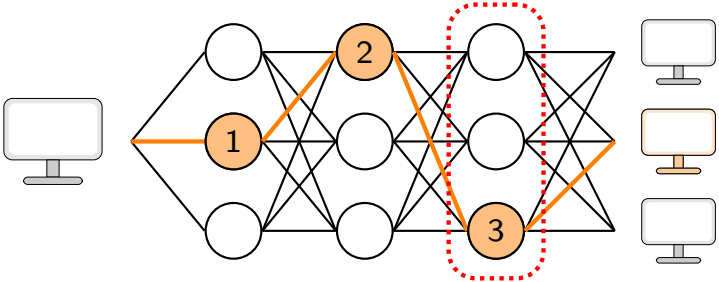


Motivation



Problem: Legitimacy of destination is only verifiable at the exit node

Motivation



Problem: Legitimacy of destination is only verifiable at the exit node

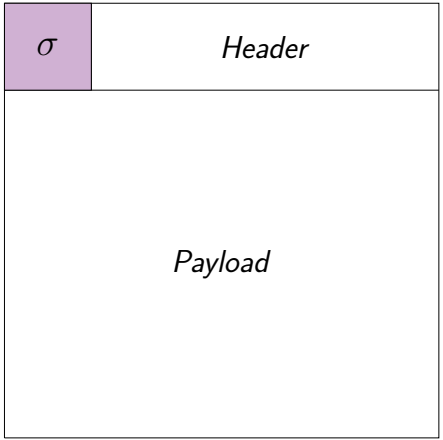
Objective: Legitimacy of destination verifiable at any hop without compromising privacy

Objective

Idea: Add a credential σ to the packet's header

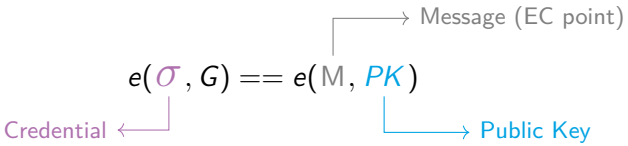
Desired properties:

- Linked to the final destination
- Verifiable at any hop
- Preserve unlinkability
  I.e. updated at each hop
- Preserve client anonimity
  I.e. cannot be linked to the client
- Preserve client privacy
  I.e. do not leak client information



General Idea

In general, credentials are verified using bilinear pairing such as:



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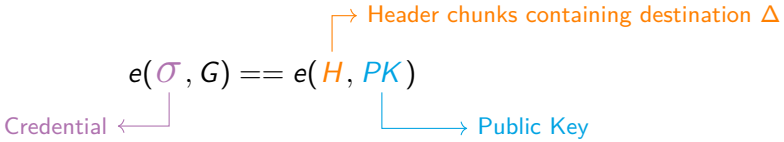
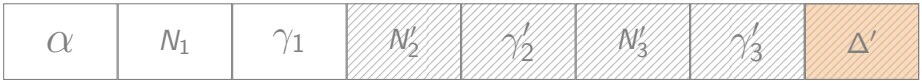


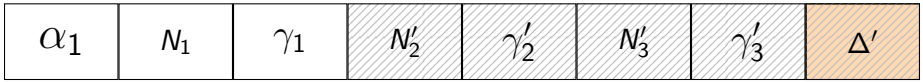
Header:



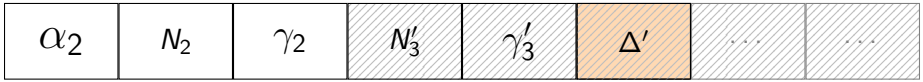
General Idea

Problem: When packet is processed, the final destination chunk Δ is shifted.

Header **at node 1**:



Header **at node 2**:



Header **at node 3**:

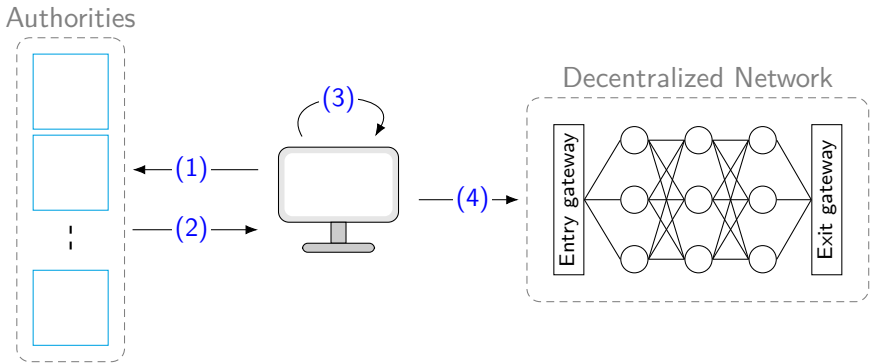
General Idea

Solution: Using all chunks that could contain final destination Δ

$$e(\sigma, G) == e(N_2' + N_3' + \Delta', PK)$$

Header:

Multi-authorities System Architecture



(1) Request
(2) Issue
(3) Aggregate
(4) Rerandomize credential

Assumption: Multi-authorities setup

Credential Construction & Update

Credential Construction:

s_i = shared secret *client* - *mixnode*$_i$

$$\sigma_1 = y\,(\Delta + N_3 + N_2 + (s_1 + s_2 + s_3)G_1 + (s_1 + s_2)G_3 + (s_1)G_5)$$

Credential Construction & Update

Credential Construction:

s_i = shared secret *client* - *mixnode*$_i$

$$\begin{aligned}\sigma_1 &= y\,(\Delta + N_3 + N_2 + (s_1+s_2+s_3)G_1 + (s_1+s_2)G_3 + (s_1)G_5) \\ &= \underbrace{y\Delta}_{\text{Credential given by authority}} + \underbrace{yN_3}_{\text{Public Parameters}} + \underbrace{yN_2}_{\text{Public Parameters}} + (s_1+s_2+s_3)\underbrace{yG_1}_{\text{Public Parameters}} + (s_1+s_2)\underbrace{yG_3}_{\text{Public Parameters}} + (s_1)\underbrace{yG_5}_{\text{Public Parameters}}\end{aligned}$$

Credential Construction & Update

Credential Construction:

s_i = shared secret *client* - *mixnode*$_i$

$$\begin{aligned}\sigma_1 &= y\,(\Delta + N_3 + N_2 + (s_1+s_2+s_3)G_1 + (s_1+s_2)G_3 + (s_1)G_5) \\ &= \underbrace{y\Delta}_{\text{Credential given by authority}} + \underbrace{yN_3 + yN_2 + (s_1+s_2+s_3)yG_1 + (s_1+s_2)yG_3 + (s_1)yG_5}_{\text{Public Parameters}}\end{aligned}$$

Credential Update at each hop:

$$\sigma_{i+1} = \sigma_i - s_i(yG_1 + yG_3 + yG_5 + yG_7) - yN_{i+1}$$

Implementation & Early Results

Implementation

- **Language**: Python 3.14
- **Library**: `mclbn256`
- **Architecture**: Multi-process
- **Communication**: UDP sockets

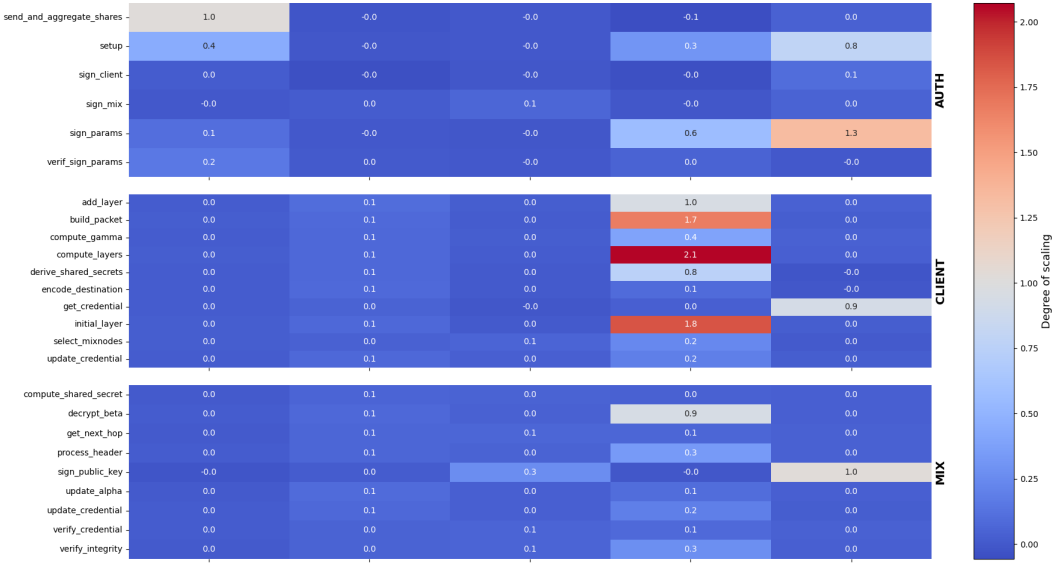
```
[16:37:34.360], CLIENT, 1, <--, AUTH, 9, G1, fc3a0d74, SIGN-CLIENT, , , 1
[16:37:34.361], CLIENT, 1, <--, AUTH, 16, G1, 88cbef0f, SIGN-CLIENT, , , 1
[16:37:34.372], CLIENT, 1, , , , , , get_credential, CPU: 18.260216 ms, wall:
[16:37:34.373], CLIENT, 1, , , , , , encode_destination, CPU: 0.619651 ms, wal
[16:37:34.373], CLIENT, 1, , , , , , select_mixnodes, CPU: 0.050984 ms, wall:
[16:37:34.375], CLIENT, 1, , , , , , derive_shared_secrets, CPU: 1.974942 ms,
[16:37:34.377], CLIENT, 1, , , , , , update_credential, CPU: 1.167153 ms, wall
[16:37:34.379], CLIENT, 1, , , , , , compute_gamma, CPU: 0.267620 ms, wall: 0.
[16:37:34.379], CLIENT, 1, , , , , , initial_layer, CPU: 2.754870 ms, wall: 2.
[16:37:34.380], CLIENT, 1, , , , , , compute_gamma, CPU: 0.212643 ms, wall: 0.
[16:37:34.380], CLIENT, 1, , , , , , add_layer, CPU: 0.853612 ms, wall: 0.857
[16:37:34.381], CLIENT, 1, , , , , , compute_gamma, CPU: 0.194506 ms, wall: 0.
[16:37:34.381], CLIENT, 1, , , , , , add_layer, CPU: 0.816702 ms, wall: 0.818
[16:37:34.382], CLIENT, 1, , , , , , compute_gamma, CPU: 0.141348 ms, wall: 0.
[16:37:34.382], CLIENT, 1, , , , , , add_layer, CPU: 0.752025 ms, wall: 0.758
[16:37:34.383], CLIENT, 1, , , , , , compute_gamma, CPU: 0.140052 ms, wall: 0.
[16:37:34.383], CLIENT, 1, , , , , , add_layer, CPU: 0.758154 ms, wall: 0.760
[16:37:34.383], CLIENT, 1, , , , , , compute_layers, CPU: 6.122563 ms, wall: 6
[16:37:34.383], CLIENT, 1, , , , , , build_packet, CPU: 10.466467 ms, wall: 10
[16:37:34.383], CLIENT, 1, -->, MIX, 1, Header, ca8a1445, HEADER, , , 1
[16:37:34.409], CLIENT, 1, <--, MIX, 34, Header, 523e5f3e, HEADER, , , 1
```

Figure: Log files

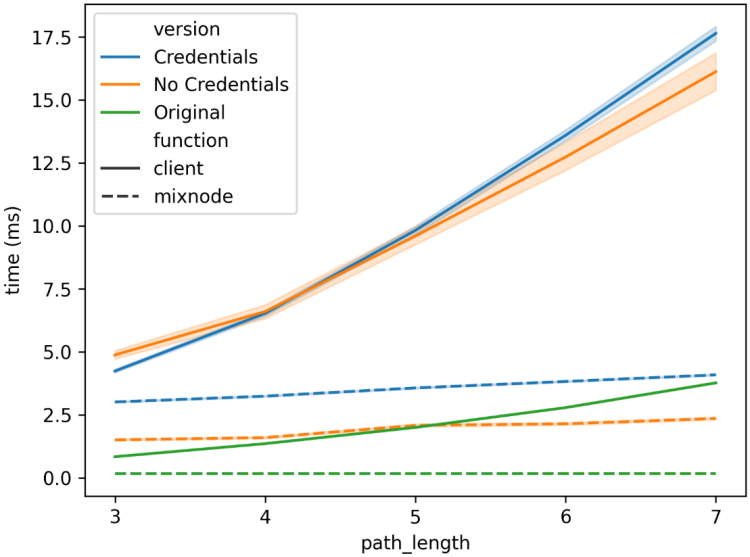
```
AUTH     8 -->  CLIENT  1       G1  6d6a909a  SIGN-CLIENT, ,
AUTH     1 -->  CLIENT  1       G1  bb44a407  SIGN-CLIENT, ,
AUTH     7 -->  CLIENT  1       G1  6b528491  SIGN-CLIENT, ,
AUTH    14 -->  CLIENT  1       G1  796cb3ad  SIGN-CLIENT, ,
AUTH    18 -->  CLIENT  1       G1  3489d2da  SIGN-CLIENT, ,
AUTH     9 -->  CLIENT  1       G1  fc3a0d74  SIGN-CLIENT, ,
AUTH    16 <--  CLIENT  1       G1  b433586d  SIGN-CLIENT, ,
CLIENT   1 <--  AUTH   11      G1  a8e8418c  SIGN-CLIENT, ,
AUTH    16 -->  CLIENT  1       G1  88cbef0f  SIGN-CLIENT, ,
CLIENT   1 <--  AUTH   15      G1  c39306c1  SIGN-CLIENT, ,
CLIENT   1 <--  AUTH   17      G1  61c7ffa2  SIGN-CLIENT, ,
CLIENT   1 <--  AUTH    7      G1  6b528491  SIGN-CLIENT, ,
CLIENT   1 <--  AUTH    8      G1  6d6a909a  SIGN-CLIENT, ,
CLIENT   1 <--  AUTH    1      G1  bb44a407  SIGN-CLIENT, ,
CLIENT   1 <--  AUTH   14      G1  796cb3ad  SIGN-CLIENT, ,
CLIENT   1 <--  AUTH   18      G1  3489d2da  SIGN-CLIENT, ,
CLIENT   1 <--  AUTH    9      G1  fc3a0d74  SIGN-CLIENT, ,
CLIENT   1 <--  AUTH   16      G1  88cbef0f  SIGN-CLIENT, ,
CLIENT   1 -->  MIX     1  Header  ca8a1445  HEADER, ,
MIX      1 <--  CLIENT  1  Header  932c6c88  HEADER, ,
MIX      1 -->  MIX    49  Header  a39e2b98  HEADER, ,
MIX     49 <--  MIX     1  Header  ef93a60b  HEADER, ,
MIX     49 -->  MIX    24  Header  dcdae061  HEADER, ,
MIX     24 <--  MIX    49  Header  41d8ffec  HEADER, ,
MIX     24 -->  MIX     2  Header  7d1a5109  HEADER, ,
MIX      2 <--  MIX    24  Header  86223ad7  HEADER, ,
MIX      2 -->  MIX    34  Header  718cdd76  HEADER, ,
MIX     34 <--  MIX     2  Header  d5f40b35  HEADER, ,
MIX     34 -->  CLIENT  1  Header  6dc10cbd  HEADER, ,
CLIENT   1 <--  MIX    34  Header  523e5f3e  HEADER, ,
```

Figure: Script running

Results: Scaling degree on CPU time execution for each functions



Results: Latency Comparison with Sphinx Baseline



Future Work

- Investigate security properties and guarantees
- Finalize the performance benchmarks
- Enhance credential capabilities (e.g. destination blacklisting, validity periods, ...)

Related Work

1. DSphinx - A decentralized Sphinx packet construction (using STTPs)
2. Decentralizing Path sampling for DSphinx (using STTPs)
3. Anonymous Credentials for DSphinx (using Multi-Authorities)